GG-A-684

MAY 22, 1947

FEDERAL STANDARD STOCK CATALOG

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

ARTICULATORS; DENTAL

This specification was approved on the above date by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government, and shall become effective not later than August 15, 1947, for use in procurement initiated on or after that date. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS

A-1. The following Federal Specification, of the issue in effect on date of invitation for bids, shall form a part of this specification.

GG-I-526—Instruments, Dental and Surgical; General Specifications.

A-2. For specifications and publications applicable only to individual departments, see section H.

B. TYPES AND CLASSES

B-1. Dental articulators covered by this specification shall be furnished in the following types and classes, as specified in the invitation for bids:

Type I.—Stephan, model "M".
Type II.—Gysi-Simplex.
Class A.—Articulator only.
Class B.—Articulator outfit, complete.
Type III.—Crown and bridge.
Type IV.—Adjustable anatomical.

C. MATERIAL AND WORKMANSHIP

C-1. *Material.*—The material shall be as specified hereinafter.

C-2. *Workmanship.*—The workmanship shall be first class throughout. The articulators shall be symmetrically made, properly finished, and free from all defects which detract from their appearance or which may impair their serviceability.

D. GENERAL REQUIREMENTS

D-1. *Instructions.*—Complete and detailed operating instructions shall be furnished with each articulator.

D-2. *Marking.*—Each articulator shall be permanently marked, in a suitable place, with the name or registered trade-mark of the manufacturer.

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D-3. *Plating and finish.*—Unless made of a relatively noncorroding metal or alloy, all parts shall receive an electroplate of metal or metals as hereinafter specified. All smooth parts shall be polished.

E. DETAIL REQUIREMENTS

E-1. *Type* I.—

E-1a. *Design.*—Shall be of the type and design known to the dental profession as Stephan articulator, model "M", as illustrated in figure 1 (A).

E-1a(1). *Upper member.*—The upper member shall be as illustrated in figure 1 (B). Unless otherwise specified in the invitation for bids, the hinge bar retaining screws shall conform to the requirements of Federal Specification GG-I-526, grade 1, class B. The flange shall be cast brass, and the knurled nuts on the ends of the hinge bar shall be wrought brass, unless otherwise specified. The complete upper member shall be nickel-plated.

E-1a(2). *Lower member.*—The lower member shall be as illustrated in figure 1 (C). The adjusting screw and thumb nut shall be brass. The corrosion-resistant steel spring shall conform to the requirements of Federal Specification GG-I-526, grade 2, class A. The flange and frame shall be cast brass. The complete lower member, with the exception of the spring, shall be nickel-plated.

E-2. *Type II.*—

E-2a. *Design.*—

E-2a(1). *Class A.*—The Gysi-Simplex articulator shall conform to the type and design illustrated in figure 2 (A), of which the individual parts are identified in figure 2 (B).

E-2a(2). *Class B.*—The articulator outfit shall conform to the type and design illustrated in figure 2 (B) plus two extra upper model supports and two extra lower model supports, one of which shall have a 10° incisor guide incline.

E-2b. *Material.*—All parts of the articulator shall be nickel-plated brass, except the following:

Straight incisor guide pin shall be nickel-plated cold-rolled steel.
Spring on the articulator frame, giving tension to the upper bow frame, shall be japanned spring steel.
Condyle screws, opening axis pins and nuts, lock screws, and incisor guide shall be nickel-plated brass.
Other screws shall be iron, with or without nickel plating.

E-3. *Type III.*—

E-3a. *Design.*—Type III articulator shall be of the type and design known to the dental profession as crown and bridge articulator, as illustrated in figure 3, and shall be suitable for accurately reproducing anatomical movements.

E-3a(1). *Upper and lower members.*—Upper and lower members shall be brass, of approximately 0.0625-inch thickness, and shall be properly reinforced along outer edges to provide adequate strength and rigidity.

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E–3a(2). *Upper member.*—The upper member shall be hinged with a corrosion-resisting steel pin, to permit opening and closing, with provisions for protrusive and lateral excursions. Antero-posterior excursion shall be not more than 30°; side excursions shall be not more than 15° with provisions to enable change of lateral excursions for working and balancing side, corresponding to the region and side of arch for which the restoration is to be constructed.

E–3a(3). *Locking screw.*—A suitable locking screw shall be provided on the back of the upper member for maintaining securely the established or desired space between the upper and lower members.

E–3a(4). *Set-lever.*—A set-lever with locking device shall be provided on the back of the articulator to retain adjustment at the following positions indicated by letters A, R, and L:

A—To cause protrusive and right and left lateral excursions to harmonize with the anatomy.

R—To create protrusive and right and left lateral excursions to coincide with the cusp inclines on the working and balancing side.

L—To change the lateral excursions to follow the cusp inclines on the working and balancing side.

E–4. *Type IV.*—

E–4a. *Design.*—The anatomical articulator shall conform to the type and design illustrated in figures 4 (A) and 4 (B). It shall be adjustable to protrusive and right and left lateral excursions, with adjustable incisal guide and guide pin at front, and two constrained guides for antero-posterior and lateral excursions. The adjustable guiding elements shall be made of steel plated with silver having a thickness of 0.001 inch, and shall contain rotatable and slidable balls of suitable metal through which corrosion-resisting steel shafts attached to the upper member are movable by sliding. The adjustable guiding elements shall be rotatably contained in two uprights, one at each end and rear of the base member, with suitable calibrations and locking device. The two uprights shall be rotatably adjustable in the base member with calibrations at their base and having a suitable locking device.

E–4a(1). *Protective plates.*—Surface exposed to plaster shall be protected by suitable metal plates cast into the upper and lower members which are to contain permanently attached but movable locking devices for retention of mounting rings.

E–4a(2). *Locking devices for guiding elements.*—Provisions to lock the articulator in centric relation and to restrict its movements to opening and closing only, shall be permanently attached to the adjustable guiding elements. The locking devices shall act upon the two shaft extremities and when disengaged shall permit protrusive and lateral excursions without interference, regardless of adjustments.

E–4a(3). *Biteforl.*—The bitefork shall be cast aluminum alloy, and one shall be furnished with each articulator.

E–4a(4). *Mounting plates.*—Mounting plates shall be aluminum alloy, and four pairs shall be furnished with each articulator.

E–4a(5). *Incisal pin extension.*—The incisal pin extension shall be cast aluminum alloy, with a suitable thread on one end for attachment to the front of the upper member.

E–4a(6). *Face bow.*—A suitable face bow shall be furnished either as a component item or as a separate item, as specified in the invitation for bids.

F. METHODS OF SAMPLING AND INSPECTION

F–1. *Sampling.*—At least one complete articulator of each type and class shall be taken at random from each lot delivered for inspection.

F–2. *Inspection.*—Samples shall be carefully inspected for material, workmanship, design, dimensions, finish, markings, and other applicable requirements.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging.*—Unless otherwise specified, commercial packages are acceptable.

G–2. *Packing.*—Unless otherwise specified, the subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to the point of delivery called for in the contract or purchase order.

G–3. *Marking.*—

G–3a. *Packages.*—Each cardboard box or carton shall be plainly labeled with the name of the item, quantity contained, and the name of the manufacturer or contractor.

G–3b. *Shipping containers.*—Unless otherwise specified, the shipping containers shall be marked with the name of the item, type, class, and quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, the number of the contract or order, and the gross weight.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS

H–1. The following specifications and publication, of the issues in effect on date of invitation for bids, and special requirements, shall form a part of this specification, and shall be applicable to purchases made under this specification by the respective departments.

H–2. *Army and Navy.*—

H–2a. *Applicable specifications and other publications.*—

Joint Army-Navy Specifications:

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

JAN–P–116—Packaging and Packing for Overseas Shipment—Preservation, Methods of.

JAN–P–120—Packaging and Packing for Overseas Shipment—Cartons, Folding, Paperboard.

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

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JAN-P-133—Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.
JAN-P-140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

U. S. Army Specification:
100-2—Standard Specification for Marking Shipments By Contractors.[1]

Navy Department Specifications:
General Specifications for Inspection of Material.[2]

Bureau of Supplies and Accounts Publication:
Navy Shipment Marking Handbook.[2]

H-2a(1). Copies of Federal and Joint Army-Navy specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department. Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Federal, Joint Army-Navy, and U. S. Army specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H-2a(2). Copies of Federal and Joint Army-Navy specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H-2b. *Packaging.*—

H-2b(1). *Individual packages.*—Articulators shall be individually packaged in a snug-fitting, folding, paperboard carton conforming to the requirements of Joint Army-Navy Specification JAN-P-129, or in a set-up paperboard box conforming to the requirements of Joint Army-Navy Specification JAN-P-133; each carton or box shall be suitably taped or glued to prevent accidental opening. Each unit shall be packaged in accordance with method IA of Joint Army-Navy Specification JAN-P-116, except that no special cleaning or preservation shall be required; technique shall conform to method IA-2 and appendix IV.

H-2b(2). *Carrier cartons.*—Twenty-four articulators, of type III only, shall be further packaged in a suitable carrier carton.

H-2c. *Packing.*—

H-2c(1). *For domestic shipment.*—Unless otherwise specified, the following designated quantities for each type and class of articulator shall be packed in a nailed wood box of appropriate size

[1] Applicable to Army purchases only.
[2] Applicable to Navy purchases only.

conforming to the requirements of Joint Army-Navy Specification JAN-P-106, style 2, 2½, or 4, type 2 load (except that the requirements for container manufacturer's identification shall not apply); closure and strapping shall conform to the requirements of the appendix thereto.

Type and class:	*Number of units to a shipping container*
Type I.—Stephan articulator, model "M"	48
Type II.—Gysi-Simplex articulator:	
Class A.—Articulator only	24
Class B.—Articulator outfit, complete	24
Type III.—Crown and bridge articulator	96
Type IV.—Adjustable anatomical articulator	12

H-2c(2). *For overseas shipment.*—Unless otherwise specified, the quantities designated in paragraph H-2c(1) shall be packed in a nailed wood box conforming to the requirements of Joint Army-Navy Specification JAN-P-106, style 2, 2½, or 4, type 2 load (except that the requirements for container manufacturer's identification shall not apply); closure and strapping shall conform to the requirements of the appendix thereto. Each box shall be lined with a sealed, water-proof barrier bag, or its equivalent, made from material conforming to the requirements of Joint Army-Navy Specification JAN-P-125 for case liners; seams and closures shall be sealed with water-resistant adhesive conforming to the requirements of Joint Army-Navy Specification JAN-P-140.

H-2d. *Marking.*—

H-2d(1). *Cartons and barriers.*—Each package, carrier carton, and barrier shall be marked with the stock numbers, nomenclature, unit, quantity, and name of manufacturer.

H-2d(2). *Shipping containers.*—

H-2d(2)a. *Army.*—Shipping containers shall be marked in accordance with the requirements of U. S. Army Specification No. 100-2, and, in addition, shall bear the Joint Army-Navy and Army stock numbers. The legend "METHOD IA PKG" shall be marked on the shipping container when items are processed as indicated in paragraphs H-2b and H-2c(2).

H-2d(2)b. *Navy.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the requirements of the latest issue of the Navy Shipment Marking Handbook, and shall bear the Joint Army-Navy stock number. The legend "METHOD IA PKG" shall be marked on the shipping container when items are processed as indicated in paragraphs H-2b and H-2c(2).

H-2e. *Ordering information.*—Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packed and shipping containers marked for domestic or overseas shipment. (See pars. H-2c and H-2d.)

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I. NOTES

I–1. Purchasers should exercise any desired options offered herein and should specify type and class required. (See pars. B–1, E–1a(1), E–4a(6), G–1, G–2, G–3b, H–2c(1), and H–2c(2).)

I–2. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with the requirements of this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I–3. Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specification, or which are commercially available, but are intended to cover the types, etc. which are suitable for Federal Government requirements.

I–4. An Index of Federal Specifications may be purchased as noted in the paragraph below, price to be obtained from the Superintendent of Documents.

I–5. Copies of this specification and of GG–I–526 may be obtained upon application accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price 5 cents each.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoeverever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

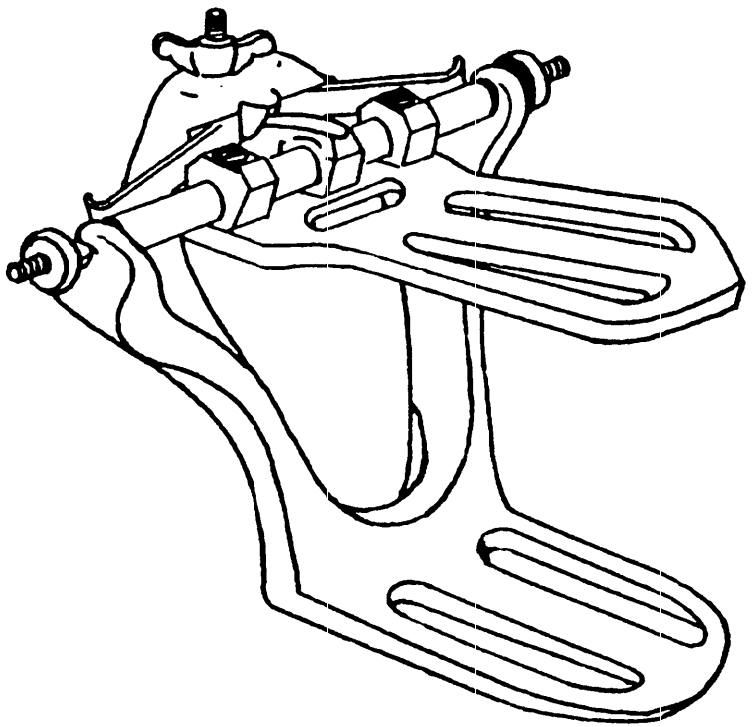


FIGURE 1 (A).—*Type I, Stephan model "M" articulator, complete.*

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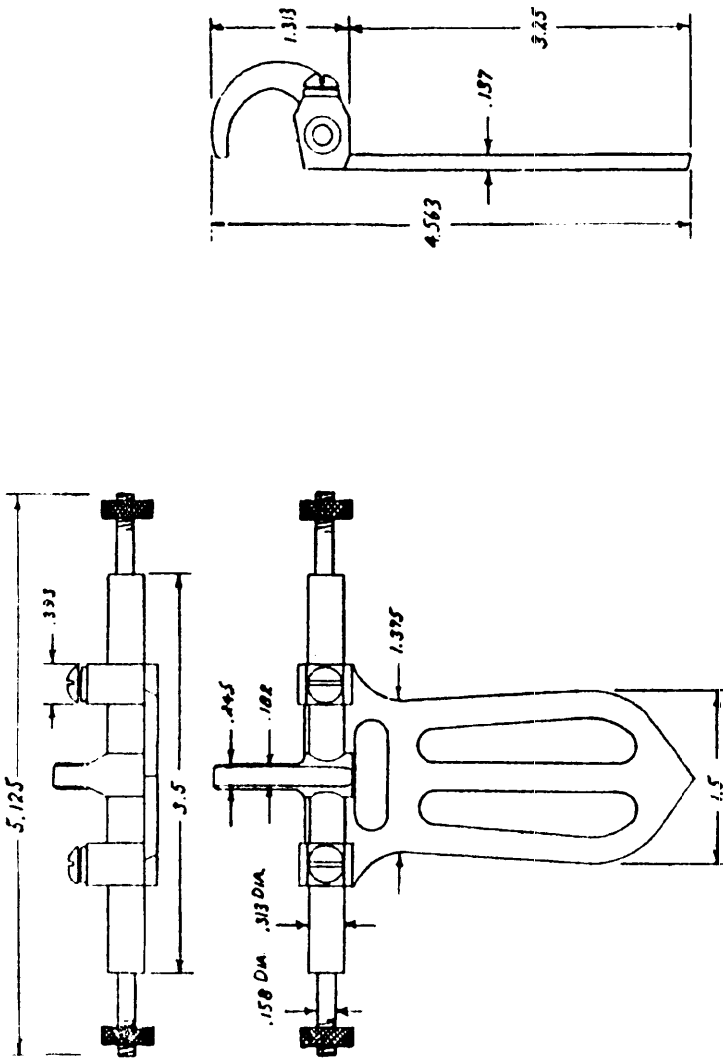


FIGURE 1 (B).—*Upper member of type I articulator.*

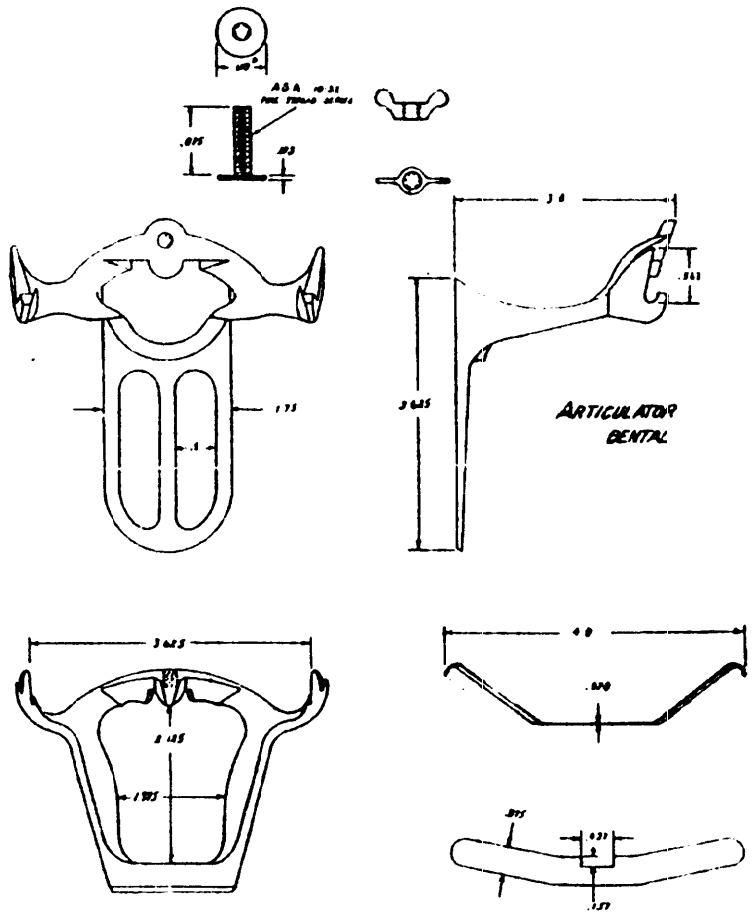


FIGURE 1 (C).—*Lower member of type I articulator.*

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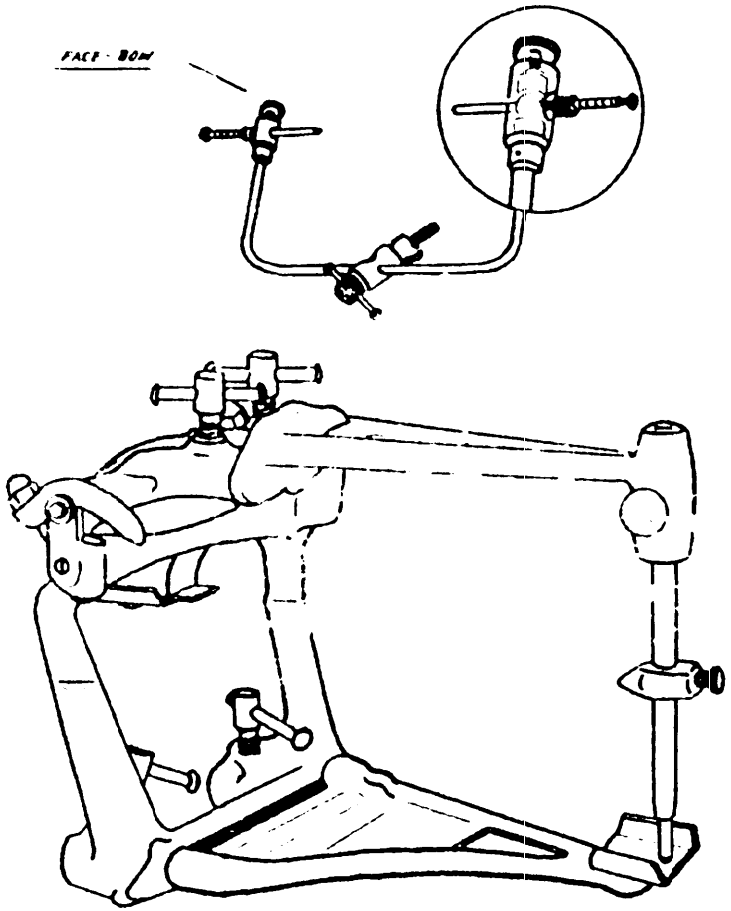
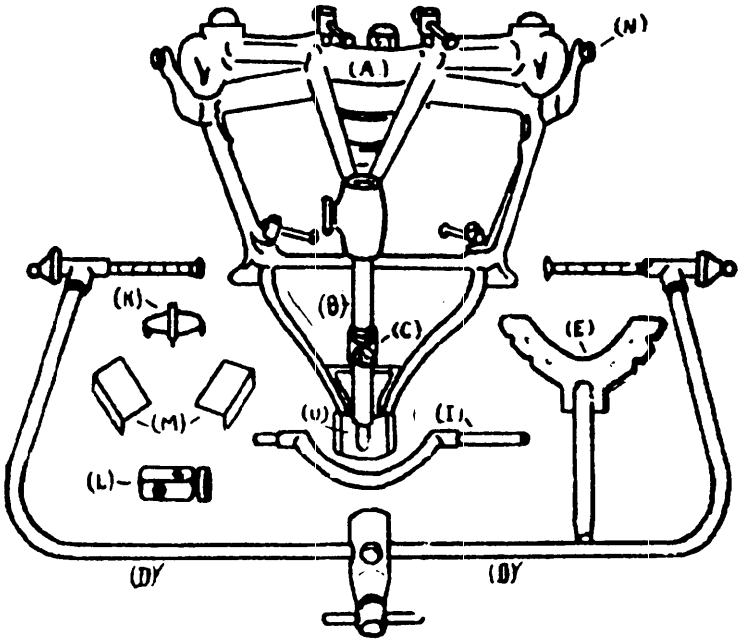


FIGURE 2 (A).—*Type II, class A, Gysi-Simplex articulator.*



A—Gysi-Simplex articulator. (See fig. 2 (A).)
B—Straight incisor guide pin.
C—Incisor guide.
D—Snow face bow.
E—Incisor path plate and holder, offset type.
I—Curved incisor guide pin.
K—Incisor path marker.
L—Anterior supporting block.
M—Bite-lock trays.
N—Articulator frame, of which face-bow adapters are a part.
J—Incisor guide incline.

FIGURE 2 (B).—*Type II, class B, Gysi-Simplex articulator outfit, complete.*

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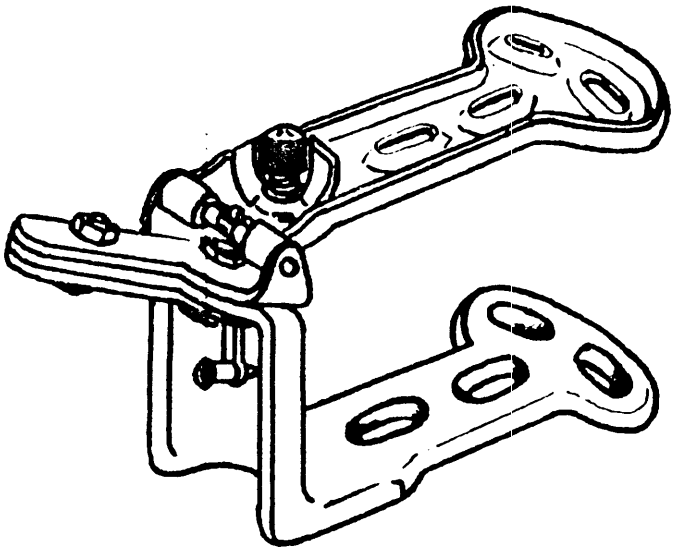
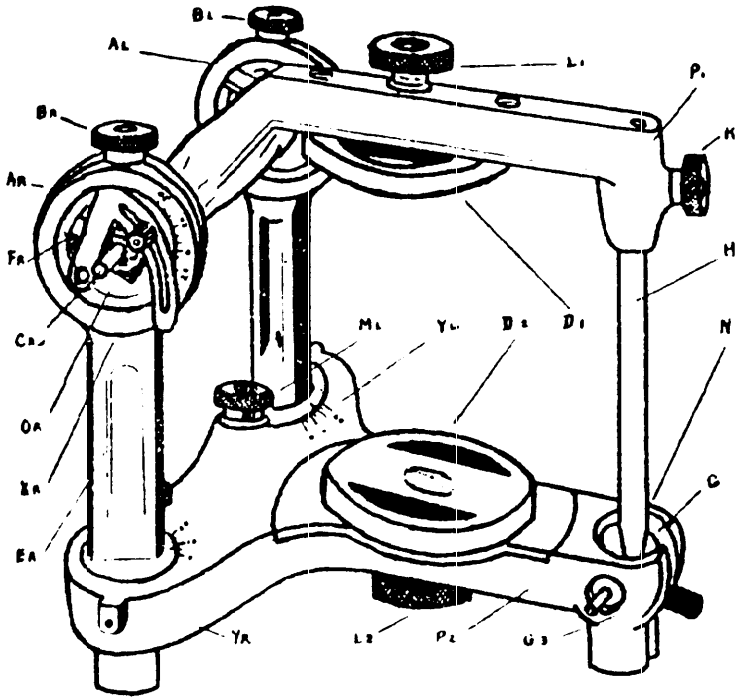


FIGURE 3.—*Type III, crown and bridge articulator.*



Al and Ar—Left and right horizontal condylar guides.
Bl and Br—Locknuts for left and right horizontal condylar guide adjustments.
Cl* and Cr—Left and right terminals of condylar shaft.
D_1 and D_2—Upper and lower mounting plates.
El* and Er—Left and right condylar posts.
Fl* and Fr—Left and right centric locks and locknuts.
G—Incisal guide.
G_r—Handle for incisal guide.
H—Incisal pin.
K—Lockscrew for incisal pin.
L_1 and L_2—Lockscrew for upper and lower mounting plates.
Ml and Mr*—Locknuts for left and right lateral adjustments.
N—Locknut for incisal guide adjustment.
Ol* and Or—Left and right condylar head (ball) elements.
P_1 and P_2—Upper and lower jaw members.
Xl* and Xr—Left and right calibration for horizontal adjustments.
Yl* and Yr—Left and right calibrations for lateral adjustments.

* Not shown because of position of sketch.

FIGURE 4 (A).—*Type IV, adjustable anatomical articulator.*

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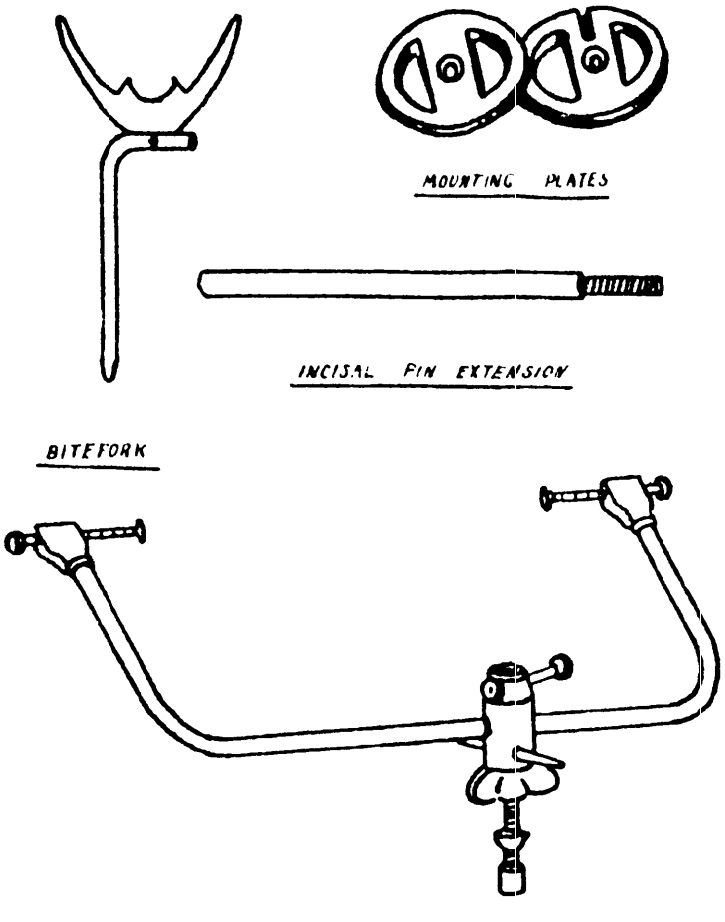


FIGURE 4 (B).—*Accessories for type IV articulators.*

For sale by the Superintendent of Documents, Washington 25, D. C.
Price 5 cents.

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